C-E-002193(Army-GL)
October 6, 1983

INTERIM FEDERAL SPECIFICATION

EGG LOAF WITH CHEDDAR CHEESE, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the US Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers egg loaf with cheddar cheese thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
JJJ-S-30	-	Salad Oil, Vegetable
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 8940

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification For

Military Standard:

MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

U.S. Department of Agriculture (USDA)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Application for copies should be addressed to the Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

Cheeses and Related Cheese Products

Standard of Identity for Cream Cheese

U.S. Standards for Grades of Cheddar Cheese

(Application for copies should be addressed to Director, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Cheese Institute

Cheese Color Standards

(Application for copies should be addressed to the National Cheese Institute, 110 North Franklin Street, Chicago, IL 60606.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.2.1 Whole eggs. The whole eggs may be liquid or frozen and shall have been processed in accordance with Regulations Governing the Inspection of Eggs and Egg Products. The whole eggs shall have been prepared from egg yolks and egg whites in their natural proportions as broken directly from the shell egg as evidenced by a USDA Egg Products Inspection Certificate. The whole eggs shall contain no additives, and shall be in satisfactory condition, free of objectionable odors such as musty, sour, or fruity, and free from foreign material. Liquid egg products shall be held at a temperature of 40°F or lower and shall not be held more than 72 hours from time of pasteurization until the start of formulation of the egg loaf. Frozen whole eggs shall be held at 10°F or lower and used within 120 days. Thawed, previously frozen, egg products shall be held at a temperature of 40°F or lower and shall not be held for more than 24 hours prior to use in processing.

3.2.2 Cheese, cheddar. The cheese shall have a well-defined medium sharp, natural cheese flavor and shall comply with or exceed the U.S. Grade A requirements of U.S. Standard for Grades of Cheddar Cheese. The cheese shall be shredded and shall be free from rind, rind rot, cheese pests and mold. The color shall be a medium yellow-orange (No. 5 - No. 8 of the National Cheese Institute's Cheese Color Reference Standard).

3.2.3 Cream cheese, fresh. The fresh cream cheese shall meet the Standard of Identity for Cream Cheese (21 CFR 133.133). The cream cheese shall have a delicate butterlike or creamy flavor, shall be moderately salted, shall have a comparatively light texture, reasonably firm body, no free whey, and shall be uniformly creamy-white in color. It may have a very slight, clean, acid flavor, but shall be free from bitter, sour, flat, yeasty, or moldy flavors. The body and texture shall not be coarse, crumbly, excessively smooth, gummy, grainy, sticky, overfirm, or too soft. The cream cheese shall be in excellent condition, free of mold, slime, discoloration, or burnt particles.

3.2.4 Salad oil. Salad oil shall conform to type B of JJJ-S-30.

3.2.5 Microcrystalline cellulose. Microcrystalline cellulose shall be a gel forming cellulose composed of 85 parts microcrystalline cellulose and 15 parts sodium carboxymethylcellulose. It shall be a white, water dispersible powder which shall retain less than 0.1 percent on a standard 60 mesh screen and less than 50 percent on a standard 325 mesh screen and have an average particle size of 28 microns in dispersion. The microcrystalline cellulose shall have an ash content of not more than 2 percent and a heavy metals content of less than 10 parts per million. The pH of a 2-percent dispersion shall be not less than 6 nor more than 8 and shall have an acid demand requiring 2.2 milliliters of 0.1 normal hydrochloric acid to change the pH of 100 milliliters of a 1 percent solids dispersion to a pH of 4. The microcrystalline cellulose shall be insoluble in water and organic solvents, but shall be dispersible in water to form colloidal sols and gels.

3.2.6 Salt. Salt shall be non-iodized, white, refined, sodium chloride with or without anti-caking agents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.2.7 White pepper. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. Volatile oil content shall be not less than 1 mL per 100 g.

3.2.8 Citric acid. The citric acid shall be Food Chemicals Codex grade.

3.3 Preparation and processing. The egg loaf shall be prepared from the following ingredients:

Ingredient	Percent by weight
Whole eggs, pasteurized	59.00
Cheddar cheese, shredded	18.20
Water	15.80
Cream cheese	4.00
Salad oil	1.60
Microcrystalline cellulose	0.80
Salt	0.50
White pepper, ground	0.05
Citric acid	0.05

a. The microcrystalline cellulose shall be combined with 75 percent of the water and mixed in equipment that provides a high degree of shear, such as homogenizers, vertical-cutter mixers or other similar equipment. Planetary mixers, lightening mixers, or similar equipment shall not be used. Proper dispersion of microcrystalline cellulose in water shall be demonstrated by a pudding-like (gel-like) consistency and flow on a viscosimeter of 1.5 cm at 70°F in 3 minutes.

b. The cream cheese, oil, remaining water, shredded cheddar cheese, and dry ingredients shall be mixed together. This mixture and the eggs shall be alternately blended with the microcrystalline cellulose gel at a slow speed (failure to thin the microcrystalline cellulose gel gradually will result in gel lumps which will not break up). In addition, final mixing of the product shall be at a slow speed.

(Note: High speed mixing will incorporate air into the product, causing air pockets in the end item. The presence of large air pockets may cause rejection of the lot.)

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the following finished product requirements:

a. The average net weight shall be 100 ounces minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

b. No individual can shall contain less than 98 ounces of product.

Each tray pack can shall be hermetically sealed under a vacuum of not less than 10 inches nor more than 12 inches of mercury immediately after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to glass, paint, metal, dirt, insect, insect parts, wood, or paper.

b. There shall be no foreign odor or flavor such as, but not limited to rancid, scorched, burnt, stale, sour, musty, moldy, or sulfurous.

c. There shall be no foreign color such as, but not limited to purple, blue, green or red.

d. The product shall possess a typical cooked egg color.

e. There shall be not more than four gas or air pockets larger than 0.3 inch in any dimension.

f. There shall be not more than 1 ounce of free liquid in any can.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting it to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, US Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 100 ounces or if the net weight of any individual can is less than 98 ounces, the lot shall be rejected.

4.6.2 Product inspection. Eight tray packs shall be selected at random from the lot. The eight tray packs shall be heated for 15 minutes in 140° to 150°F water, opened, and inspected in accordance with table I.

a. If no defects are found, the lot shall be accepted.

b. If two or more defects are found, the lot shall be rejected.

c. If only one defect is found, a second sample of eight tray packs shall be selected at random from the lot, heated as above, and inspected for the defects listed in table I. If any defect, is found in the second sample, the lot shall be rejected.

TABLE I. Product inspection 1/ 2/

Category	Defect
Major	
151	The presence of more than four air pockets larger than 0.3 inch in any dimension. 3/
152	The presence of more than 1 ounce (by weight) of free liquid in a tray pack can. 3/
153	Cheese particles not evident.
154	Presence of a mild to strong sulfur odor.
155	Product not possessing a typical cooked egg color.

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.) foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall only be performed when deemed necessary by an AMS agent).

3/ To determine the weight of free liquid and to inspect for gas and air pockets, the can shall be opened and with the lid held in place the free liquid shall be decanted into a tared container allowing 1 minute for draining to occur. The remaining product shall be divided into ten approximately equal portions and examined for gas and air pockets.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used; (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above the top edge of box.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 100 ounces of the food product, sealed and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width and be fabricated of grade V3c fiberboard. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure sensitive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width and be fabricated of the same material as the box.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. Paper labels are not permitted. Cans shall show the following statements:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 35 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 20 servings per tray, cut 4 slices in width by 5 slices in length.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included.

CAUTION: DO NOT STACK PALLETS IN TRANSIT,
OR MORE THAN 2 HIGH IN STORAGE

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8910-A430

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY
US Army Natick Research and Development Laboratories
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRDNA-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Laboratories
ATTN: DRDNA-ES
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
C-E-002193(Army-GL)	Egg Loaf With Cheddar Cheese, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP. 1 APR 83

Source: https://assist.dla.mil – Downloaded: 2026-10-28T11:05Z
Check the source to verify that the content is current before use.